

Voice Emergency Checklist System

Generic advisory reader · driven by the installed Aircraft Card

One combined reference: product, Aircraft Card spec, certification basis, hardware, and enclosure. Supersedes the separate tech, processor, and enclosure documents.

Cert path: NORSEE (PS-AIR-21.8-1602) · DO-160G environmental · deliberate DO-178C-avoidance argument · STC/PMA placement.

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS

Not airworthy and not certified. The certification sections describe the path a productized unit would follow — a plan, not held approvals. No NORSEE LOA, STC, PMA, or TSOA exists for this article.

Scope & Status

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS. This document describes a bench / training prototype and a certification-readiness roadmap. As built, the unit is not airworthy, not certified, and must never be installed in or relied upon aboard an aircraft. Nothing here is an STC, TSO authorization, PMA, or NORSEE letter of approval. The certification sections (Part C) describe the path a productized unit would follow — they are a plan, not a claim of compliance.

The product is a generic, offline, voice-driven advisory emergency-checklist reader. The pilot states an emergency by name; the device reads each checklist item aloud and waits for a spoken completion word before advancing. It runs entirely on-device (no cloud, no Wi-Fi). The product itself is aircraft-agnostic — all aircraft-specific behavior comes from the Aircraft Card that is installed (Part B). The Cessna Citation CJ2 is included only as the reference example card.

This master reference supersedes and combines the previously separate documents: the technical data packet, the processor-selection note, and the enclosure specification. Those remain in the repository for history; this is the single source of truth.

Part	Contents
A Product	What the system is, the generic architecture, and the safety model
B Aircraft Card	The card-defines-everything model + the formal card specification & validation
C Certification basis	NORSEE / DO-160G / installation path, and the deliberate DO-178C-avoidance argument
D Hardware reference	Pin map, per-device wiring, annunciator, lamp driver, power, BOM, processor selection
E Enclosure	Two-piece mechanical specification for the fabricating engineer
F References	All cited regulatory and component sources

PART A

The Product

A generic, offline, advisory checklist reader. This part defines what the system is (and is not), its block-level architecture, and the three safety rules that the entire certification argument in Part C is built on.

A.1 What it is (and is not)

It is	It is not
An advisory read-aloud reader of checklist items	A required or primary aircraft system
Independent — no electrical/data tie to any aircraft system	An interface to avionics, engines, or controls
Driven entirely by an installed Aircraft Card	Tied to one airframe in firmware
Offline, deterministic, single-chip	A cloud / connected / large-vocabulary STT device
A complement to the certified/required checklist	A substitute for the AFM/QRH or required checklist

This framing is not cosmetic — it is the foundation of the certification argument in Part C. A device that is non-required, advisory-only, independent of primary systems, and fails to a clearly-annunciated safe state is the textbook profile for the NORSEE (Non-Required Safety Enhancing Equipment) approval path, and it is what lets the program lean on DO-160G environmental qualification while avoiding DO-178C software assurance.

A.2 Generic system architecture

Block	Function	Aircraft-specific?
MCU + speech stack	Wake word → command recognition → playback sequencing	No — fixed firmware
Microphone (I2S)	Captures crew speech for recognition	No
Speaker + amplifier (I2S)	Reads checklist items / annunciations aloud	No
microSD (the Aircraft Card)	Carries aircraft ID, checklist library, audio, config, validation	Yes — the only aircraft-specific element
Annunciator switch	Dark-cockpit status / fault indication, IN/OUT select	No

The MCU runs Espressif ESP-SR (AFE noise-suppression/VAD → WakeNet wake word → MultiNet fixed-grammar command recognition). The grammar (trigger phrases, advance words) is small and bounded, which is what keeps an MCU-class part in scope (see D.7).

A.3 The safety model (carried into the cert argument)

Three design rules define the failure behavior, and each one maps to a NORSEE requirement (Part C):

- 1 Revert-to-unopened. If the card is missing, unreadable, malformed, fails schema validation, or any referenced audio clip is absent, the device refuses to present any checklist and enters a

clearly-annunciated FAULT state. It never shows partial or stale data. (*Failure mode = loss of function, not misleading information.*)

- 2 Dark-cockpit annunciation. When selected IN and healthy, the unit shows nothing. A fault lights the amber FAULT legend. Selected OUT shows the white OFF status and inhibits the fault legend (a deliberately deselected system needs no crew action). A power-up lamp test proves the legend is alive.
- 3 Independence. The device draws power and nothing else from the aircraft; it neither reads from nor writes to any aircraft system. Loss of the device cannot affect any primary function.

PART B

The Aircraft Card

Card defines everything. The product is generic; the installed microSD card makes it a specific aircraft's checklist reader. This part is the formal card specification — layout, manifest schema, voice-grammar rules, and the validation logic that implements revert-to-unopened.

B.1 Principle

The product is generic. The installed card makes it a specific aircraft's checklist reader. To support a new airframe — King Air, PC-12, TBM, another Citation — you author a new card. No firmware change, no recompile, no hardware change.

Everything that varies by aircraft lives on the card: the aircraft identity, the checklist library (titles, trigger phrases, ordered items, per-item completion/advance words), the read-aloud audio, and the universal advance vocabulary. The firmware contains only the generic engine that loads, validates, and plays whatever a valid card provides.

B.2 Card layout (microSD, FAT32)

```
/sdcard/
  config.txt           (optional) one line: AIRCRAFT=
  /                   one folder per aircraft (e.g. CJ2, B200, PC12)
    checklists.json    the card manifest + checklist library
    audio/
      .wav             one read-aloud clip per referenced item + fault clips
```

- Auto-select: if exactly one aircraft folder is present, it loads automatically.
- Explicit select: if multiple folders exist, config.txt names the active one. A missing/invalid target is a fault (revert-to-unopened), not a silent guess.

B.3 Card manifest schema (checklists.json)

Top-level object:

Field	Type	Req.	Meaning
schema_version	integer	rec.	Card schema version the firmware validates against
aircraft	string	yes	Short aircraft id; must match the folder name
title	string	yes	Human-readable aircraft name shown in logs
universal_advance	string[]	yes	Advance words accepted on every item (e.g. check, checked, complete, next)
checklists	object[]	yes	The checklist library (≥ 1)

Each checklists[] entry:

Field	Type	Req.	Meaning
id	string	yes	Stable identifier (e.g. engine_fire)
title	string	yes	Spoken/displayed checklist name
type	string	rec.	emergency, abnormal, normal
triggers	string[]	yes	Phrases that invoke this checklist by voice
items	object[]	yes	Ordered checklist steps (≥ 1)

Each items[] entry:

Field	Type	Req.	Meaning
clip	string	yes	Audio basename in audio/ (read-aloud <clip>.wav)
text	string	yes	The item text (for logs / optional display)
advance	string[]	opt.	Item-specific completion words (in addition to universal_advance)

B.3.1 Voice-grammar rules (validation-enforced)

MultiNet (English) imposes grammar constraints that the card author must honor; the validator rejects a card that violates them (revert-to-unopened):

- Phrases are lowercase letters and single spaces only — no digits or punctuation.
- Spell numbers as words: “V-one” → v one, not V1.
- Keep the total command count modest (well under the ~200-command MultiNet cap).
- Every clip referenced by any item must exist as audio/<clip>.wav.

B.4 Card validation & the revert-to-unopened rule

On boot the firmware attempts a complete, valid load and returns exactly one status:

Status	Meaning	Annunciation
STORE_OK	SD mounted, JSON parsed + schema-valid, every audio clip present	dark (healthy)
FAULT_NO_CARD	SD not detected / mount failed	amber FAULT
FAULT_NO_AIRCRAFT	No aircraft folder, or config.txt target missing	amber FAULT
FAULT_NO_JSON	checklists.json missing / unreadable	amber FAULT
FAULT_PARSE	JSON malformed	amber FAULT
FAULT_VALIDATION	Schema or voice-rule violation, empty data	amber FAULT
FAULT_AUDIO_MISSING	A referenced clip is absent	amber FAULT
FAULT_NO_MEMORY	Allocation failed while loading	amber FAULT

On any FAULT the in-memory checklist table is left EMPTY — the device cannot present a partial or stale checklist even if asked. This is the literal implementation of “always revert to unopened if files are not available.”

B.5 Card integrity & provenance (recommended for a productized card)

For a card that drives a *safety-enhancing* device, integrity matters as much as schema validity.

Recommended additions (forward-looking, beyond the current demo):

- Schema version gate — firmware refuses a card whose `schema_version` it does not support, rather than mis-parsing it.
- Manifest checksum / signature — a per-card hash (and, for production, a signature) so a corrupted or tampered card is rejected as `FAULT_VALIDATION`.
- Content provenance — record, per card, the AFM/QRH source revision the checklist text was transcribed from, the author, and the date. Advisory equipment is only as good as the source it mirrors; provenance is part of the configuration-control story the FAA expects under NORSEE (Part C).
- Read-only media — distribute production cards write-protected.

PART C

Certification Basis

STC / PMA / NORSEE-ready reference. This part is a roadmap, not evidence of compliance — the demo unit holds no approvals. It explains why NORSEE is the right path, the standards elected as Minimum Design Requirements, and the deliberate, documented argument for avoiding DO-178C.

Status: this part is a roadmap, not evidence of compliance. The demo unit holds no approvals. The intent is to design and document so that a productized unit can pursue approval efficiently.

C.1 Why NORSEE is the right path

The product profile — non-required, advisory, independent of primary systems, failing to a clearly-annunciated safe state — matches the FAA's Non-Required Safety Enhancing Equipment (NORSEE) policy, PS-AIR-21.8-1602 (issued 03/31/2016)¹. NORSEE approval is a combined design and production approval issued under 14 CFR § 21.8(d) for equipment that is “not required by any Federal regulation with the intent to measurably increase aircraft safety,” determined to be a minor change to type design with a minor failure condition. A voice reader that simply reads aloud the same items already in the certified/required checklist — while the paper/electronic required checklist remains the authority — is a strong fit.

C.1.1 Applicability limit (read this)

NORSEE policy applies to aircraft certified under 14 CFR Part 23, 27, and 29 (and predecessor categories) and explicitly excludes Part 25 transport-category aircraft¹.

- The reference example, the Citation CJ2 (525A), is a Part 23 airplane — so NORSEE is available for it.
- Because the product is generic, the card library will inevitably include Part 25 aircraft (e.g. larger jets). For a Part 25 installation, NORSEE is not available; that installation must use an STC (or other major/minor-change path appropriate to the airframe). The product's certification status is therefore per-airframe, and the documentation must say so explicitly.

C.2 Minimum Design Requirements (the standards we elect)

Under NORSEE the applicant proposes the industry standard(s) that become the Minimum Design Requirements (MDR); the FAA accepts, partially accepts, or augments them. The FAA recommends widely-accepted RTCA / SAE / ASTM standards¹.

Topic	Elected standard / basis	Notes
Environmental	RTCA/DO-160G	Primary qualification evidence. Categories in C.4. FAA accepts DO-160 D/E/F/G per AC 21-16G and “strongly encourages DO-160G for new articles.”
Software	No DO-178C sought — see C.3	Argued out via function/failure classification, not avoided by omission
Complex hardware	DO-254 not invoked	COTS MCU + simple discrete logic; no custom complex devices (ASIC/FPGA/PLD). Per AC 20-152A, <i>simple</i> hardware verifiable by test needs no DO-254 design assurance.

Topic	Elected standard / basis	Notes
Human factors / color	FAA AC 25-11B conventions	amber=caution, white=status, dark-cockpit. Accepted design convention even though the unit is not a Part 25 article.
System safety (if needed)	SAE ARP4761 / AC 23.1309-1	Only invoked if the safety evaluation ever rises above "minor" (it should not)

C.3 The DO-178C–avoidance argument (deliberate, documented)

DO-178C is software *design-assurance* guidance whose rigor scales with the failure condition the software can contribute to. NORSEE lets us classify the function and, for a genuinely minor–failure advisory device, scope software assurance down to essentially nothing formal — provided the architecture earns it. The argument has four legs:

- 1 Function is advisory, not required. The device does not perform, command, or feed any required function. The crew's authority is the certified/required checklist; this unit only reads it aloud.
- 2 Worst-case failure is minor. The two credible failures are (a) loss of function — it goes silent or annunciates FAULT, and the crew uses the paper/QRH exactly as today; and (b) misleading information — mitigated by the revert-to-unopened rule (no partial/stale output), card validation, and a procedural requirement that the crew cross-checks against the required checklist. Neither failure reduces the crew's ability to cope with a condition worse than minor — the NORSEE safety-evaluation test¹.
- 3 Independence. No input from or output to any primary system; physical and electrical separation. This is one of the design considerations the policy lists for keeping a failure minor¹.
- 4 Qualitative safety evaluation is permitted for non-complex equipment; a quantitative probabilistic analysis (and the DO-178C machinery that feeds it) is not required for a minor–failure advisory function¹.

Honest caveats (must stay in the doc): (1) This argument must be agreed with the FAA ACO early — the applicant proposes the classification; the FAA concurs. If the FAA judges the failure condition above minor (e.g. because of over-reliance / automation-dependency human factors), the program moves to §2 of the NORSEE policy (xx.1309, ARP4754A/ARP4761) and software assurance re-enters. (2) “No DO-178C” is earned by architecture and procedural mitigations, not by labeling. The revert-to-unopened behavior, the validation gate, the dark-cockpit annunciation, the independence, and a mandatory limitation that the unit may not be used as a substitute for the required checklist are the price of that classification.

C.4 DO-160G environmental qualification plan

Categories a cockpit/avionics-bay unit would target. For the prototype these are design targets, not completed tests².

DO-160G section	Target category	Rationale for this unit
§4 Temperature & Altitude	Cat A2 (controlled/pressurized)	–15 °C to +55 °C operating; pressurized cabin
§5 Temperature Variation	Cat B	Cockpit rate of change
§6 Humidity	Cat A	Conformal coat recommended
§7 Operational/Crash Shock	Operational + crash-safety	Boards retained; nothing becomes a projectile
§8 Vibration	Cat S (fixed-wing)	Locking hardware; conformal coat; no press-fit-only parts
§15 Magnetic Effect	Class Z	Small device; classify by measured deflection
§16 Power Input	per installation (e.g. 28 VDC)	Only if a 28 V variant is built; TVS + fuse
§17 Voltage Spike	Cat A	Input transient protection
§19 Induced Signal Susceptibility	Cat ZC	Cockpit

DO-160G section	Target category	Rationale for this unit
§21 RF Emission	Cat M (or better)	Wi-Fi/BT disabled in firmware materially helps emissions
§22 Lightning Induced Transient	as installed	Behind-panel mounting reduces exposure
§25 ESD	per §25	Bond exposed metal; recessed connectors
§26 Flammability	UL94 V-0 materials	Required if not aluminum

The deliberate choice to disable Wi-Fi and Bluetooth in firmware is both a security decision and an emissions-qualification advantage (§21).

C.5 Approval & installation path (per airframe)

NORSEE approval is design + production approval — not installation approval. The full chain for a Part 23/27/29 airframe:

- 1 Pre-application — engage the responsible FAA ACO early; agree on the NORSEE classification (minor change / minor failure) and the MDR (C.2).
- 2 Design + test — complete DO-160G qualification, the safety evaluation, human-factors assessment, and the configuration-control / card-provenance package.
- 3 NORSEE Letter of Approval (LOA) under § 21.8(d), with the certifying statement of compliance (template in C.6) and a quality system per the policy.
- 4 Installation — NORSEE *eligibility* still requires an installation approval on each aircraft: if a minor alteration, a logbook entry / FAA Form 337 as applicable; if a major change to type design (panel structure, electrical integration), an STC or field-approval path.
- 5 Part 25 aircraft — skip NORSEE; pursue STC for the installation (C.1.1).

C.5.1 Where TSO and PMA fit (and don't, here)

- TSO authorization is a *minimum-performance* design+production approval against a specific TSO. There is no TSO that defines a “voice advisory checklist reader,” so a TSOA is not the natural path; NORSEE (applicant-proposed MDR) is. A TSOA would also still require separate installation approval³.
- PMA is for *modification/replacement* articles for a type-certificated product and is typically tied to an STC or identity. This unit is added equipment, not a replacement part, so PMA is not the primary path — though a production unit may end up holding PMA in conjunction with an STC for specific airframes (14 CFR § 21.303).

C.6 Certifying statement of compliance (template)

Per the NORSEE policy, the LOA application carries a statement in this form (placeholders to be completed at application). *Reproduced as a template only — not a current certification.*

“I, (authorized representative), certify that (company) has complied with all applicable requirements as identified in RTCA/DO-160G and the other Minimum Design Requirements accepted for this article, and policy statement PS-AIR-21.8-1602, and that the article is produced under the required quality system.”

Two limitation statements the FAA expects on advisory NORSEE, to be placarded / in the manual:

“No operational credit may be taken for installation of this system.” · “This system is not a required system and may not be used as a substitution for the certificated aircraft checklist.”

C.7 Compliance summary matrix

Requirement	Means of compliance	Status (demo)
Non-required, safety-enhancing	NORSEE PS-AIR-21.8-1602, § 21.8(d)	Argued; not applied

Requirement	Means of compliance	Status (demo)
Minor change / minor failure	Qualitative safety evaluation (C.3)	Drafted; ACO concurrence pending
Environmental	RTCA/DO-160G (C.4) per AC 21-16G	Targets defined; not tested
Software assurance	DO-178C not sought — advisory/minor (C.3)	Architecture supports it; ACO pending
Complex hardware	DO-254 not invoked (simple COTS) — AC 20-152A	N/A by design
Human factors / color	AC 25-11B conventions, dark-cockpit	Implemented in design
Installation	Minor alteration or STC, per airframe	Per-aircraft; none performed
Part 25 airframes	STC (NORSEE excluded)	Flagged

¹ FAA Policy Statement PS-AIR-21.8-1602, "Approval of Non-Required Safety Enhancing Equipment (NORSEE)" — [gajsc.org](https://www.faa.gov/gajsc)

² RTCA/DO-160G; FAA AC 21-16G acceptance of DO-160 D-G — [faa.gov AC 21-16G](https://www.faa.gov/aircraft/air_cert)

³ FAA Technical Standard Orders (TSO) program — [faa.gov/aircraft/air_cert](https://www.faa.gov/aircraft/air_cert)

PART D

Hardware Reference

The generic hardware platform shared by every aircraft card: MCU, pin map, per-device wiring, annunciator, lamp driver, power budget, processor selection, firmware notes, and the DIY bill of materials. Pin numbers match firmware/main/board_pins.h.

Pin numbers match firmware/main/board_pins.h. The wiring diagram is in §D.10. Logic level is 3.3 V (the ESP32-S3 is not 5 V tolerant on GPIO).

D.1 System overview

Item	Value
MCU	ESP32-S3 (dual-core LX7 @ 240 MHz) — PSRAM required by ESP-SR
Recommended module	ESP32-S3-WROOM-1 N16R8 (16 MB flash, 8 MB octal PSRAM)
Speech stack	Espressif ESP-SR: AFE (NS/VAD) → WakeNet “Hi ESP” → MultiNet English
Mic input	I2S MEMS microphone on I2S_NUM_0
Audio output	I2S Class-D amplifier on I2S_NUM_1 → 4–8 Ω speaker
Config storage	microSD — the Aircraft Card (Part B), FAT32
Annunciation	Applied Avionics split-legend switch (dark-cockpit, AC 25-11B)

Two build paths: Integrated (ESP32-S3-Korvo-2 dev board — on-board dual mic, ES8311 codec, NS4150 amp, microSD slot; best mic performance) or DIY (ESP32-S3 DevKitC-1 N16R8 + INMP441 mic + MAX98357A amp + microSD breakout + the annunciator switch).

D.2 Master pin map

Function	Macro	GPIO	Dir	Notes
Push-to-talk	PTT_GPIO	0	in (PU)	BOOT button; active-low
SELECT switch	SELECT_GPIO	10	in (PU)	IN = GPIO→GND (active-low)
Legend OFF (white)	LEGEND_OFF_GPIO	21	out	top legend half (via driver)
Legend FAULT (amber)	LEGEND_FAULT_GPIO	14	out	bottom legend half (via driver)
Status LED	STATUS_LED_GPIO	48	out	on-board RGB on most S3 devkits
Mic bit clock	MIC_BCLK_GPIO	4	out	I2SO BCLK → mic SCK
Mic word select	MIC_LRCLK_GPIO	5	out	I2SO WS → mic WS
Mic data in	MIC_DIN_GPIO	6	in	mic SD → ESP DIN
SD clock	SD_CLK_GPIO	7	out	SDMMC CLK
SD command	SD_CMD_GPIO	9	i/o	SDMMC CMD (needs pull-up)

Function	Macro	GPIO	Dir	Notes
SD data 0	SD_D0_GPIO	8	i/o	SDMMC DAT0 (needs pull-up)
Speaker bit clock	SPK_BCLK_GPIO	15	out	I2S1 BCLK → amp BCLK
Speaker word select	SPK_LRCLK_GPIO	16	out	I2S1 WS → amp LRC
Speaker data out	SPK_DOUT_GPIO	17	out	I2S1 DOUT → amp DIN

Polarity macros: PTT_ACTIVE_LOW=1, SELECT_ACTIVE_LOW=1, LEGEND_OFF_ACTIVE_HIGH=1, LEGEND_FAULT_ACTIVE_HIGH=1, STATUS_LED_ACTIVE_HIGH=1, LAMP_TEST_MS=2000. Set any unused output to -1 to disable it cleanly.

Reserved / avoid pins (N16R8): GPIO33–37 (octal PSRAM/flash bus — do not use), GPIO19/20 (USB D–/D+), GPIO0/45/46 (strapping — must boot in the right state), GPIO26–32 (SPI flash on some modules). GPIO0 here is only the BOOT/PTT button, so it is safe.

D.3 Per-device wiring

D.3.1 INMP441 MEMS mic → ESP32-S3 (I2S_NUM_0). Supply 1.8–3.3 V (never 5 V), ~2.2–2.5 mA. VDD→3V3, GND→GND, SCK→GPIO4, WS→GPIO5, SD→GPIO6, L/R→GND (left channel). Decouple 0.1 μF; 100 kΩ pulldown on SD; never clock with VDD off.

D.3.2 MAX98357A Class-D amp → ESP32-S3 (I2S_NUM_1). Supply 2.5–5.5 V; ~2.4 mA quiescent; peak ~650 mA at 5 V/4 Ω; no MCLK. VIN→5 V (full output), GND→GND, BCLK→GPIO15, LRC→GPIO16, DIN→GPIO17, GAIN NC = 9 dB, SD/mode float = mono. OUT+/OUT- are bridge-tied — never to GND.

D.3.3 microSD (the Aircraft Card) → SDMMC 1-bit. 3.3 V card. CLK→GPIO7, CMD→GPIO9 (10 kΩ→3V3), DAT0→GPIO8 (10 kΩ→3V3), VDD→3V3, VSS→GND. FAT32; layout per Part B.

D.3.4 Discrete inputs. SELECT (GPIO10): LOW = selected IN; pull-up HIGH = OUT. PTT (GPIO0): LOW = pressed. Each is a simple SPST to GND; debounce in software.

D.4 Annunciator switch (split-legend, dark-cockpit)

State	TOP — white VOICE CHKLST OFF	BOTTOM — amber VOICE CHKLST FAULT
Selected IN, healthy	dark	dark ← true dark cockpit
Selected IN, fault	dark	amber ON
Selected OUT	white ON	dark (fault inhibited)

Power-up lamp test: both halves on for ~2 s, then dark, so a dead LED cannot masquerade as healthy. OUT inhibits fault per AC 25-11B — a deselected system needs no crew action, so only the white OFF status shows; recognition and fault monitoring are suspended while OUT.

D.5 Lamp-driver circuit (one per legend half)

An ESP32 GPIO (~20 mA default, 40 mA max; 1.5 A total chip limit) cannot drive a 28 V — or 5 V at lamp current — legend directly. Use a low-side switch per half:

```

+V_lamp (5 V or 28 V, separate rail)
|
[ legend lamp ]      <- VIVISUN legend (white or amber)
|
+----- Drain
GPIO21 --[1k]--|G  N-ch MOSFET (logic-level, e.g. 2N7002 / A03400)
(or G14)      |   or NPN (e.g. 2N2222 with base resistor)
[10k]         <- gate/base pulldown -> GND (defined OFF)
|
GND (Source) ----- common ground with ESP32

```

One driver for GPIO21 (OFF/white), one for GPIO14 (FAULT/amber). The 10 kΩ pulldown guarantees the lamp is OFF during boot/reset before the GPIO is configured (dark-cockpit integrity). For a 28 V incandescent legend, use a FET with $V_{ds} \geq 40$ V and current $\geq$ inrush. Applied Avionics VIVISUN/Korry legends come in 28 VDC, 5 VDC, 28 VAC, 5 VAC, and 115 V lamp variants. For a quick bench mock-up, substitute two 3.3 V LEDs (white + amber) with series resistors driven straight from GPIO21/GPIO14 within the ~20 mA limit.

D.6 Power budget

Rail	Loads	Typical	Peak
3.3 V	ESP32-S3 (Wi-Fi off) + mic + microSD	~80–150 mA	~250 mA (SD init / SR burst)
5 V	MAX98357A output	a few mA idle	~650 mA (5 V/4 Ω, loud)
Lamp rail	up to 2 legend halves	0 (dark)	per lamp spec (e.g. 28 V incand.)

Power from USB 5 V $\geq$ 1 A. Keep a 28 V legend supply separate from logic 5 V (grounds common only). Bulk decoupling $\geq$ 100 μ F near the amp VIN plus 0.1 μ F per device.

D.7 Processor selection

This is a narrow, bounded workload — a small fixed vocabulary (checklist triggers + universal advance words, well under the ~200-command MultiNet cap), offline, and deterministic. That is command recognition on a fixed grammar, not open-ended transcription, which keeps an MCU-class part firmly in scope and makes a Linux SBC overkill. It is also exactly the property that supports the minor-failure / no-DO-178C argument (C.3): a single-chip, fixed-grammar device has a small, well-understood failure surface.

Recommendation: keep the ESP32-S3 as the baseline; consider the ESP32-P4 only for headroom. ESP-SR (v2.1+) supports S3 and P4 for English MultiNet; the classic ESP32 is no longer supported by the current speech algorithms and should be avoided.

Option	Summary	Verdict
ESP32-S3 (baseline)	LX7 dual-core @240 MHz + AI vector ext, 512 KB SRAM, Wi-Fi+BLE, ~\$8–15	Best AI/voice part in the line, most mature tooling; firmware/diagram/BOM built on it
ESP32-P4 (upgrade)	RISC-V to 400 MHz + AI ext, 768 KB SRAM, ~2.5× compute, no Wi-Fi/BT	Worth it only for more compute or a future display; no-radio is arguably a safety plus
Syntiant NDP120	Always-on ultra-low-power NDP	Overkill; we have panel power, need full grammar + playback + SD
Picovoice Porcupine+Rhino	Offline wake+intent on Cortex-M4	Clean fit but needs a per-deployment license key — undesirable for self-contained safety gear
Fluent.ai / NXP i.MX RT600	Commercial intent / heavy audio DSP	More licensing/board complexity than a fixed grammar needs
Raspberry Pi (whisper/Vosk)	Full Linux STT	Non-deterministic boot, higher power/cost — less robust for a fixed-grammar safety device

D.8 Firmware build notes

Topic	Value
Framework	ESP-IDF $\geq$ 5.2
Components	esp-sr, esp_spiffs, driver, json (cJSON), fatfs, sdmmc, esp_driver_sdmmc
Speech models	WakeNet WN9_HIESP; MultiNet English mn6_en/mn7_en (S3 only)
Partitions	factory app 3 MB + model 5 MB + storage 2 MB $\rightarrow$ needs 16 MB flash (N16R8)
Grammar rules	lowercase + single spaces; spell numbers ("v one"); ~200-cmd cap
Fault behavior	any card fault $\rightarrow$ ST_FAULT, amber legend, no checklist shown

D.9 Bill of materials (DIY build)

Qty	Part	Spec / example
1	ESP32-S3 DevKit	DevKitC-1 N16R8 (PSRAM)
1	I2S MEMS mic	INMP441 / ICS-43434 breakout
1	I2S amp	MAX98357A breakout
1	Speaker	4–8 Ω , $\geq$ 2 W
1	microSD card + breakout	FAT32 (the Aircraft Card)
1	Annunciator switch	Applied Avionics VIVISUN/Korry split-legend (or 2 LEDs for bench)
2	Lamp driver	logic-level N-MOSFET (2N7002/AO3400) or NPN (2N2222)
4	Resistors	1 k Ω $\times$ 2 (gate), 10 k Ω $\times$ 2 (pulldown)
2–3	Pull-ups	10 k Ω on SD CMD/DAT0 (if breakout lacks them)
—	Caps	0.1 μ F per device, 100 μ F bulk near amp
1	PTT button	momentary SPST (or use BOOT)

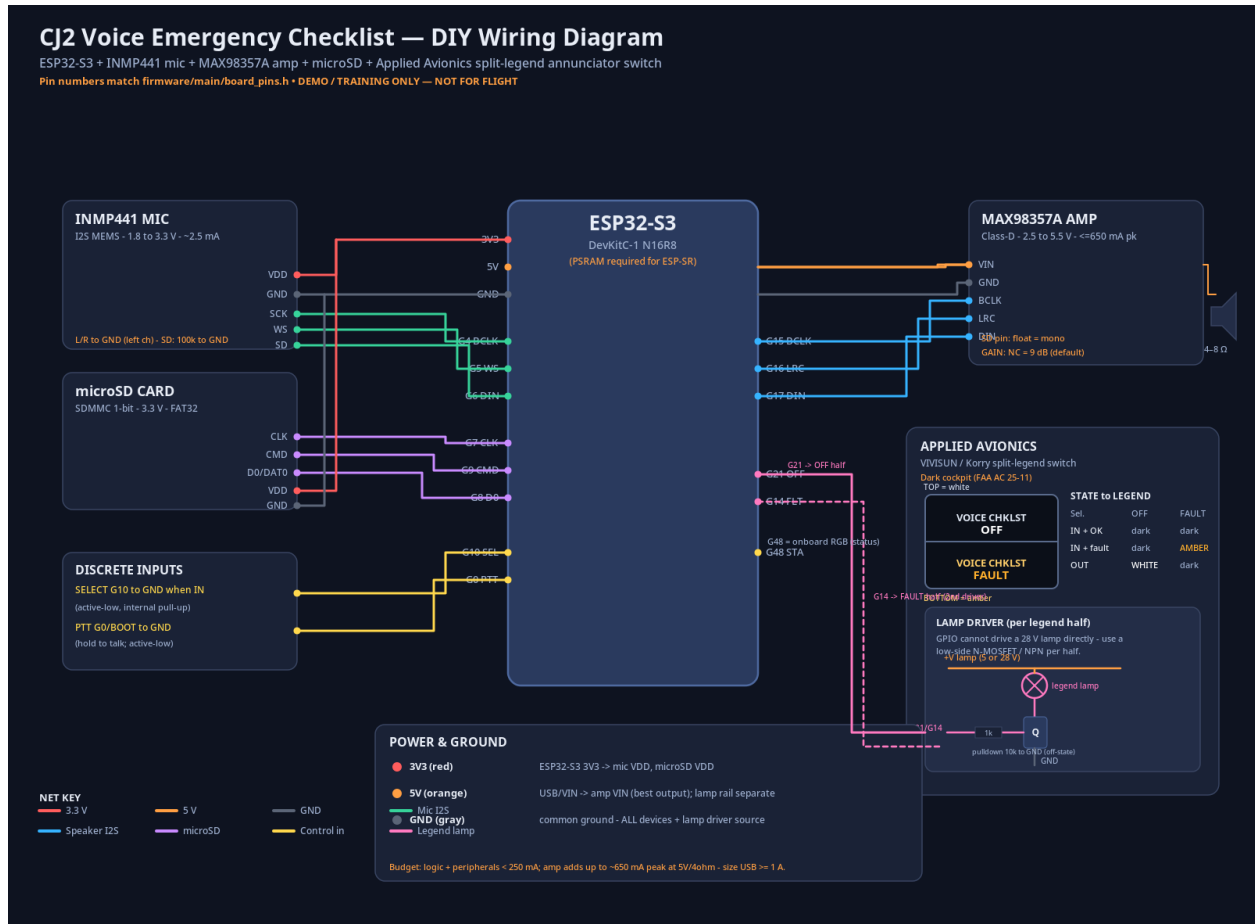
Integrated alternative: ESP32-S3-Korvo-2 (~\$45–55) replaces mic/codec/amp/SD; use its BSP pin map and ES8311 codec init.

D.10 System wiring diagram

CJ2 Voice Emergency Checklist — DIY Wiring Diagram

ESP32-S3 + INMP441 mic + MAX98357A amp + microSD + Applied Avionics split-legend annunciator switch

Pin numbers match firmware/main/board_pins.h • DEMO / TRAINING ONLY — NOT FOR FLIGHT



Complete system wiring — ESP32-S3, mic, amp, microSD (Aircraft Card), and split-legend annunciator with per-half lamp drivers.

PART E

Enclosure Specification

For the fabricating engineer. The housing is generic — sized for the electronics and the panel interface, not for a specific airframe. The only airframe-specific item is the mounting variant (DZUS slot vs. round instrument hole) and the card inside it.

E.1 Two-piece architecture

Piece	Contents	Where	Why
A. Panel bezel	Split-legend annunciator switch, speaker + grille, optional PTT	Front panel / pedestal, on the DZUS rail	Crew must see/reach it; annunciator in the normal scan
B. Remote box	ESP32-S3, mic, amp, microSD slot, lamp-driver, power conditioning	Avionics bay / behind-panel, blind	Keeps heat, the card slot, and wiring out of the panel

A single all-in-one box is acceptable for a pure bench demo, but the two-piece split mirrors real remote-mount avionics and keeps the mic away from fan/avionics noise.

E.2 Piece A — panel bezel

- DZUS rail mount: fastener pitch 3/8 in (9.525 mm); clearance hole 0.255 in (6.48 mm); bezel height a whole multiple of 3/8 in (target 3-unit = 28.575 mm, or 4-unit = 38.1 mm if the grille needs room); standard pedestal width $\approx$ 146 mm aluminum (144.45 mm face); backplate 1/16 in (1.6 mm) 6061-T6; first/last fastener 14.29 mm from each end.
- Round-hole variant: fits a standard 3-1/8 in (79.4 mm) instrument cutout with four 6-32 screws on the standard bolt circle, to replace a blanking plate where no DZUS slot exists.
- Face layout (top→bottom): split-legend annunciator switch (cut per the *specific* switch datasheet — typical VIVISUN bezel $\approx$ 15×15 mm to 19×19 mm; top half VOICE CHKLST OFF white, bottom VOICE CHKLST FAULT amber, upright when installed); speaker grille ($\geq$ 40 % open over the cone, offset from the switch); optional guarded/recessed PTT.
- Material/finish: 6061-T6 aluminum 2.0–3.0 mm (or ABS/PC for a non-structural demo); matte black, low-gloss ($\leq$ 10 gloss units) to suppress glare; legend by the switch's internal engraving (preferred) or laser-etch + white/amber paint-fill; edges chamfered 0.5 mm.

E.3 Piece B — remote processor box

- Envelope: sized around the ESP32-S3 DevKitC-1 ($\approx$ 70×26 mm) plus amp, mic, microSD breakout, and the 2-channel lamp-driver; practical outer $\approx$ 110 × 80 × 45 mm. Confirm against the actual stacked board set.
- Mounting: internal standoffs / M2.5 brass inserts — boards screwed down, not floating (vibration). Keep the mic away from the amp/any fan; if the mic lives here, add a meshed acoustic port; mic may instead live in the bezel (keep the I2S run < 150 mm).

- Access & connectors: externally swappable microSD (Aircraft Card) carrier labeled CONFIG CARD – FAT32 (swap without opening the box); covered/recessed USB-C service port (bench use only); one keyed, positive-latching main connector (small MIL-circular or 9-pin D-sub) carrying SELECT, both legend drives, PTT, speaker +/–, power/ground (pinout from board_pins.h). Accept USB 5 V $\geq$ 1 A; optional internal 28 V $\rightarrow$ 5 V DC-DC ($\geq$ 2 A) with TVS + fuse if a 28 V bus mock-up is wanted (mark as demo regulator, not DO-160 qualified).
- Material/EMI: aluminum preferred (doubles as EMI shield + heatsink); if plastic, add a grounded conductive shield liner/coating; single-point chassis ground stud bonded to the connector shell and ESP32 ground.

E.4 Audio, thermal, environmental, labeling

- Audio: 4–8 Ω , ≥ 2 W speaker, sealed-back or small rear volume (5–15 cm³); grille ≥ 40 % open with acoustic mesh; gasket the speaker to prevent buzz.
- Thermal: ESP32-S3 + ESP-SR is low-power (a few hundred mW) — no fan. Passive convection (vent slots low/high) or conduction (thermal pad to the aluminum wall). If sealed, verify internal rise < 20 °C above 55 °C ambient. DC-DC (if fitted) on its own thermal path.
- Environmental: design toward the DO-160G categories in C.4 (Cat A2 temp, Cat S vibration, etc.) — design guidance for the prototype, formal test for a productized unit.
- Labeling: placard DEMO / TRAINING ONLY — NOT FOR FLIGHT; box exterior carries unit name, serial/asset field, FAT32 card format, and the USB “bench use only” note; annunciator legends VOICE CHKLST OFF (white) / VOICE CHKLST FAULT (amber); amber = caution, white = status per AC 25–11B.

E.5 Deliverables & open items for the engineer

Deliverables: STEP + native 3D CAD of both pieces (boards + switch modeled in place); fully-dimensioned 2D drawings (DZUS pattern, switch cutout from the chosen datasheet, grille, connector cutouts; GD&T on the switch cutout and DZUS holes); connector pinout mapped to board_pins.h; an FDM/SLA printable prototype for fit-check; a mechanical BOM; tolerances (switch cutout ± 0.1 mm, DZUS holes ± 0.1 mm on the 9.525 mm pitch, general ± 0.25 mm).

Open items (confirm before CAD): the exact Applied Avionics switch part number — the single most critical dimension; nothing finalizes until it is fixed. Also: DZUS slot vs. 3–1/8 in round hole in the target panel; where the mic lives; whether a 28 V input is wanted; and the speaker model (sets grille open area + rear-volume cavity).

Reference dimensions: DZUS pitch 9.525 mm · DZUS hole 6.48 mm · backplate 1.6 mm · first fastener offset 14.29 mm · pedestal panel width ≈ 146 mm · round instrument hole 79.4 mm · remote box $\approx 110 \times 80 \times 45$ mm · speaker 4–8 Ω ≥ 2 W.

PART F

References

All cited regulatory, component, and mechanical sources, with clickable links.

Certification & regulatory

#	Reference	Link
1	FAA Policy Statement PS-AIR-21.8-1602 — “Approval of Non-Required Safety Enhancing Equipment (NORSEE)”	open
2	FAA Technical Standard Orders (TSO) program overview	open
3	FAA AC 21-16G — acceptance of RTCA/DO-160 versions D–G	open
4	RTCA DO-160G — Environmental Conditions and Test Procedures for Airborne Equipment	open
5	FAA AC 20-152A — Development Assurance for Airborne Electronic Hardware (simple-hardware relief)	open
6	FAA AC 20-168 / RTCA DO-313 — installation of non-essential, non-required equipment	open
7	TSO / TC / STC / PMA primer (Afuzion)	open
8	FAA AC 25-11B — electronic flight displays; annunciation color / dark-cockpit conventions	open

Hardware & components

#	Reference	Link
1	INMP441 microphone datasheet	open
2	MAX98357A amplifier datasheet (Analog Devices)	open
3	MAX98357A breakout guide (Adafruit)	open
4	ESP32-S3 GPIO drive current discussion	open
5	microSD operating current	open
6	Applied Avionics VIVISUN lighted pushbutton switches	open
7	Espressif ESP-SR speech framework	open
8	Espressif ESP32-P4 product page	open
9	ESP32-P4 vs ESP32-S3 performance	open

Mechanical / panel

#	Reference	Link
1	DZUS panel-building dimensions guide (MyCockpit)	open
2	Standard 3-1/8 in instrument cutout (Aircraft Spruce)	open

DEMO / TRAINING ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS. Repository: github.com/flas-tech/cj2-voice-emergency-checklist (MIT License). The certification sections describe a path, not held approvals; no NORSEE LOA, STC, PMA, or TSOA exists for this article.